

Personalized Dual Transformer Network for sequential recommendation

Meiling Ge^a, Chengduan Wang^a, Xueyang Qin^b, Jiangyan Dai^a, Lei Huang^c, Qibing Qin^{a,c} ,
Wenfeng Zhang^d

^a School of Computer Engineering, Weifang University, Weifang, China

^b School of Computer Science and Technology, Dalian University of Technology, Dalian, China

^c Faculty of Information Science and Engineering, Ocean University of China, Qingdao, China

^d College of Computer and Information Science, Chongqing Normal University, Chongqing, China

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ABSTRACT

Sequential Recommendation (SR) seeks to anticipate the item that users will interact with at the next moment, utilizing their historical sequences of interactions. Its core task is to mine users' interests. Most existing transformer-based models focus on extracting user's interests by leveraging local context information from individual interaction sequences, ignoring the user's personalized characteristics. Furthermore, when the sequence length is significantly shorter than the specified threshold, these transformer-based models encounter challenges associated with the cold-start issue. To tackle the mentioned issues, we propose a novel transformer-based sequential model, named Personalized Dual Transformer Network (PDTN), that extends the length of the user's sequences by leveraging the user's historical behaviors to search for relevant items. Specifically, a personalized feature extraction module is proposed to extract characteristics of both the user and the user's historical behaviors. Then, a Dual Transformer is designed to retrieve a user's relevant historical behaviors, which can increase the length of the user's historical interaction sequences and fully extract the user's interests. Furthermore, Stochastic Shared Embeddings (SSE) regularization is developed into a Transformer-based sequential model to reduce overfitting and errors in the processing of training. Extensive experiments are conducted to compare PDTN with current methods on four publicly available real-world datasets for prediction tasks. The results consistently demonstrate that PDTN outperforms advanced sequential recommendation methods. The source code is publicly available at <https://github.com/QinLab-WFU/PDTN>.

1. Introduction

With the exponential growth of the Internet and the advancement of technologies in collecting, processing, and analyzing vast amounts of data, we are now immersed in an era where big data plays a pivotal role [1–3]. While enjoying the convenience of big data, humans are also facing the challenge of information overload. Therefore, it is particularly important to obtain information of interest to users from massive data. Recently, many researchers have used recommendation systems to mine information that people are interested in and solve the problem of information overload [4], such as online shopping [5], digital entertainment platforms [6], etc.

Traditional recommendation systems, such as collaborative filtering algorithms, are utilized to capture consistent user–item interactions and are designed to infer users' overall preferences [7,8]. However, they neglect the order of user–item interactions, limiting their ability to effectively capture dynamic changes in user interests within historical

sequences of actions. Sequential recommendation has emerged as a new research focus compared to static traditional recommendation methods.

The task of sequential recommendation involves building a model that utilizes the sequential patterns of user–item interactions to extract their interests and then predict the next potentially exciting item for the users [9–11]. Markov Chain [12] serves as a classic model in sequence recommendation, where it assumes that the next action depends only on the previous action (or a sequence of preceding actions). SPMC [13] utilizes feedback from the sequences and social interactions within the same model to enhance the performance of sequential recommendation. Recently, some scholars have introduced deep learning technology into sequential recommendations to improve the accuracy of recommendation [14–18]. For instance, DREAM [19] uses a recurrent neural network (RNN) architecture that incorporates temporal dynamics to predict users' future baskets of goods. By merging short-term sessions with long-term behaviors, SDM [20] captures the

* Corresponding author at: School of Computer Engineering, Weifang University, Weifang, China.

E-mail addresses: gemeiling@wfu.edu.cn (M. Ge), 20111182@wfu.edu.cn (C. Wang), qinxueyang@snnu.edu.cn (X. Qin), daijy@wfu.edu.cn (J. Dai), huangl@ouc.edu.cn (L. Huang), qinbing@wfu.edu.cn (Q. Qin), itzhangwf@cqu.edu.cn (W. Zhang).

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dynamic preferences of users. Although these methods are effective in some cases, they are partially restricted to particular types of data. By leveraging strong simplifying assumptions, Markov chain-based approaches exhibit promising performance in extremely sparse contexts. However, their efficacy in capturing the intricate dynamics of more intricate scenarios is limited. In contrast, RNNs have better expressive power but typically demand substantial amounts of data to outperform simple baselines.

Recently, a novel sequential model known as Transformer has demonstrated remarkable efficiency and advanced performance in the field of machine translation [21]. The Transformer model relies exclusively on a self-attention mechanism, which has shown remarkable effectiveness in recognizing grammatical and semantic structures among words in a given sentence. At present, a lot of works have introduced Transformers into sequential recommendation [22–24]. SASRec [9] is a representative work that applies Transformers to the recommendation field. SASRec transforms a user's historical sequence of actions into a matrix, using a multi-head attention mechanism to compute similarity scores among various items. Several subsequent works [25–27] have improved the SASRec model to improve recommended performance, demonstrating the effectiveness of the Transformer in the sequential recommendation.

Although Transformer-based methods are effective, there are still several issues: (1) Lack of personalization. As shown in the upper picture of Fig. 1, several prevailing sequential models solely focus on extracting user interests by utilizing the local context information from a single interaction sequence. However, they tend to overlook the crucial aspect of capturing the user's characteristics, which is indispensable for effective recommendation. However, users' interests are not only related to their historical behaviors but also related to their characteristics such as age and gender. (2) If the sequence length is significantly shorter than the fixed length, excessive zero-padding may impair the Transformer's capability to model item associations within the sequence, potentially causing a reduction in the model's recommendation performance. Obviously, a short user sequence makes it difficult to accurately extract user's interests, resulting in an inaccurate recommendation. As shown in Fig. 1, user *a* has a short length of historical sequence, posing challenges in fully capturing the interests of the user *a*.

To tackle the above problems, this paper proposes a new sequential recommendation approach to consider personalized factors and increase sequence correlation by using a Dual Transformer to add the user's relevant historical behaviors, called Personalized Dual Transformer Network (PDTN). Specifically, our proposed PDTN framework mainly contains Personalized Feature Extraction Module and a Dual Transformer, as depicted in Fig. 2. For the Personalized Feature Extraction Module, both the features of the user and the user's historical behaviors can be captured. At the same time, we introduce the regularization of Stochastic Shared Embeddings [28] for the sake of preventing overfitting. In Dual Transformer, taking into account the varying interaction sequence lengths among different users, we initially employ the Item Enhancement module to predict potentially interesting items for users to align their sequence lengths. Subsequently, we utilize Interest Representation Learning to extract the user's interests to predict the next item.

The main contributions of this paper are as follows:

- To obtain more personalized user features, we develop a Personalized Feature Extraction Module that extracts both user-specific features and historical behavior features effectively enhancing the model's ability for personalized recommendations.
- To fully capture the user's interests, a Dual Transformer is developed to retrieve a user's relevant historical behaviors and incorporate them with the user's recent behaviors to generate the final recommendation.

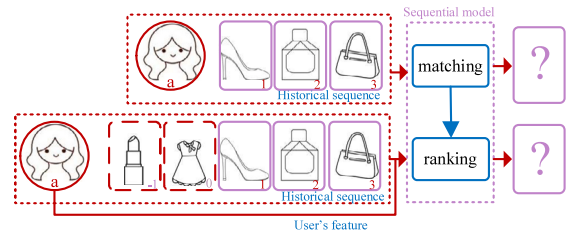


Fig. 1. A description of the motivation for the PDTN method. Simply considering the user's historical behavior cannot fully extract the user's interests. The user's purchasing interest is also related to the user's characteristics. As shown in the upper picture of Fig. 1, when the historical sequence of user *a* is very short, it is difficult to fully explore the user's interests based on the user's short sequence. Therefore, as depicted in the lower picture of Fig. 1, we not only consider the characteristics of the users themselves but also retrieve relevant behaviors based on historical behaviors and combine them with their recent behaviors to generate the final recommendations.

- To further reduce overfitting and errors in training, Stochastic Shared Embedding regularization is employed to randomly replace an embedding with another embedding with some pre-defined probability.
- Extensive evaluations on four benchmark datasets demonstrate that our proposed method achieves superior performance compared to other advanced methods.

The remainder of this paper is organized as follows. Section 2 provides an overview of relevant works in the field. In Section 3, we present a detailed description of the PDTN model. Section 4 demonstrates the effectiveness of the proposed PDTN architecture through comprehensive experiments. Finally, Section 5 summarizes our conclusions and presents our perspectives for future work.

2. Related work

This section will examine several prior studies that are closely connected to the subject matter of this paper. It mainly includes Sequential Recommendation and Attention Mechanism for Sequential Recommendation.

2.1. Sequential recommendation

The Sequential Recommendation model is designed to capture and summarize users' preferences based on their historical behaviors, enabling accurate predictions for the next item of interest. Early works on sequential recommendation are typically based on the Markov chains [13,29,30]. For instance, Rendle et al. combine matrix factorization with Markov chains to effectively capture long-term user preferences and dynamic transitions [31]. Kabbur et al. propose a method that leverages the product of two matrices to learn item similarity, enabling the generation of top-notch recommendations even in scenarios with sparse datasets [32]. A series of works follow this route and extend it to higher-order Markov chains [29,33]. However, Markov chain-based approaches typically focus on modeling sequential transitions between adjacent items. This may limit their ability to capture users' long-term preferences. Furthermore, these models may not effectively simulate the dynamic changes over time.

Due to rapid advancements in deep neural networks, researchers have recently proposed numerous methods based on RNNs for modeling sequential patterns in user's interaction sequences [34–36]. For example, Cui et al. employ a recurrent structure to dynamically capture user interests and explore a more efficient approach for handling multiview features [37]. Wu et al. present a session-based recommendation architecture that utilizes graph models to represent session sequences [38]. Wang et al. propose a deep knowledge perception network using a knowledge graph to represent news recommendations. In addition to

RNN, convolutional neural network (CNN) has also been employed to model sequential patterns between sequences. Tang et al. leverage CNNs to capture sequential patterns in user-item interactions [35]. NextItNet consists of multiple stacked convolutional layers, allowing it to effectively increase receptive fields without relying on pooling operations. This enables the model to capture long-range dependencies within the historical sequences [39]. Yan et al. encode the item embedding sequence into paired representations and utilize 2D CNN to extract sequence features [40].

To improve the performance of recommendations, multi-modal sequential recommendation methods are proposed. For example, Zhang et al. propose a scalable decomposition model that leverages visual signals as indicators of user preferences, which is applied to a selection of large real-world datasets [41]. Wang et al. propose a micro-video recommendation framework called Dual Graph Neural Network (DualGNN), which is based on a user-micro-video bipartite graph and user co-occurrence graph, utilizing user correlations to collaboratively mine personalized fusion modes for each user [42]. Meng et al. propose a heterogeneous multi-task learning framework, called “PiNet”, which learns visual features that integrate semantic and collaborative information by training an image encoder to complete ingredient prediction and food recommendation tasks simultaneously [43]. Chen et al. propose a fashion recommendation neural network architecture based on image region-level features and user review information [44]. Although these deep neural network models have performed well in sequential recommendation tasks, they exhibit suboptimal performance when applied to sparse datasets.

Recently, the success of the Transformer also inspires the development of a Transformer-based sequential recommendation model. In contrast to the Markov chain and RNN approaches, Transformer-based models utilize attention mechanisms to capture relationships between all items within sequences. To further enhance the performance, Sun et al. introduce a bidirectional Transformer layer [27]. Li et al. use the self-attention mechanism and temporal attention mechanism to model temporal relationships in user’s behavior sequences [26]. Although these Transformer-based sequence models have achieved good performance in sequence recommendation tasks. However, they often struggle to fully extract users’ interests and perform poorly on sparse data sets.

2.2. Attention mechanism for SR

The attention mechanism has proven to be effective in various tasks, including natural language processing [21], machine translation [45], and image caption generation [46]. Recently, attention mechanisms have been used in the recommendation domain to improve the recommendation performance and interpretability. For example, Zhou et al. calculate the similarity between the user’s historical behaviors and the candidate products, utilizing the attention mechanism to dynamically generate contextualized vector weights [47]. Zhou et al. employ a self-attention mechanism to encode the sequence of user’s behaviors, allowing it to effectively capture temporal dependencies between different behaviors, for instance, by capturing how earlier behaviors affect subsequent ones [48].

The above works use the attention mechanism as an addition to the original model. With the prominence of purely attention-based Transformer [21] in machine translation tasks, Transformer has also been used in recommendation systems and has achieved state-of-the-art results in sequential recommendations [49–52]. For example, Fan et al. propose a Wasserstein self-attention module to describe item-item spatial relationships in sequence, effectively incorporating uncertainty into model training to enhance model performance [23]. Liu et al. propose a non-invasive self-attention mechanism under the BERT framework to use side information to generate better attention allocation [53]. Ji et al. propose incorporating a latent space within self-attention to model the recommendation context by leveraging a multivariate biased

Table 1

Symbols and descriptions.

Symbols	Definition
U	the set of users
I	the set of items
S^u	user’s historical sequence
E^I	item embedding matrix
E^U	user embedding matrix
d_u	the latent dimension of user
d_i	the latent dimension of item
N	the maxlength of sequence
P	position embedding matrix
S_0^u	the previous item in a sequence
L	threshold value of sequence length
H	the number of enhancing item
B	the number of self-attention blocks
r_{it}	the predicted score of item i by user u at t
F_t^b	the output hidden units associated with the Transformer at the next timestamp
h	the number of attention heads

normal distribution. Their approach takes into account the covariance matrix of co-occurrence, item features, and user information [54].

However, Transformer-based models may face challenges in effectively modeling the preferences of users with short historical behavior sequences. This makes it harder to accurately capture and extract their interests. Distinct from the previously mentioned methods, this paper incorporates personalized information and strengthens the item sequence to comprehensively extract the user’s interests and generate subsequent recommendations.

3. Methodology

In this section, we first define the problem of sequential recommendation (Section 3.1) and then introduce the backbone Transformer for solving the problem (Section 3.2). Furthermore, we present the PDTN model to effectively find related items enhancing the items for next-item prediction. The overall architecture of the PDTN is shown in Fig. 2 and comprises three modules: Personalized Feature Extraction Module (Section 3.3), Dual Transformer (Section 3.4), and Prediction Layer (Section 3.5). Some symbols and descriptions commonly used in this paper are shown in Table 1.

3.1. Problem formulation

The sequential recommendation aims to generate the next recommendation by learning and summarizing users’ interests according to their characteristics and the items they have interacted with. We denote the set of users and items as U and I . For each user $u \in U$, we can express his historical sequence of interactions as follows:

$$S^u = (S_1^u, S_2^u, \dots, S_{|S^u|}^u), S_i^u \in I (i = 1, 2, \dots, |S^u|), \quad (1)$$

where $|S^u|$ denotes the length of historical interaction sequences. The goal of the sequential model is to predict the next item $S_{|S^u|+1}^u$ of interest based on the user’s historical sequence S^u , which can be represented as: $p(S_{|S^u|+1}^u | S^u)$.

3.2. Transformer for recommendation

Since we have integrated the Transformer as the fundamental component of our sequence encoder, we introduce it beforehand before outlining our model. The configuration of the Transformer is illustrated on the left-hand side of Fig. 2, consisting of two essential elements: a multi-head self-attention module and a position-wise feed-forward network (FFN). The rationale behind the Transformer is that items within sequences exhibit correlations, but their relevance to other items at different positions within the sequence may vary. Specifically, when

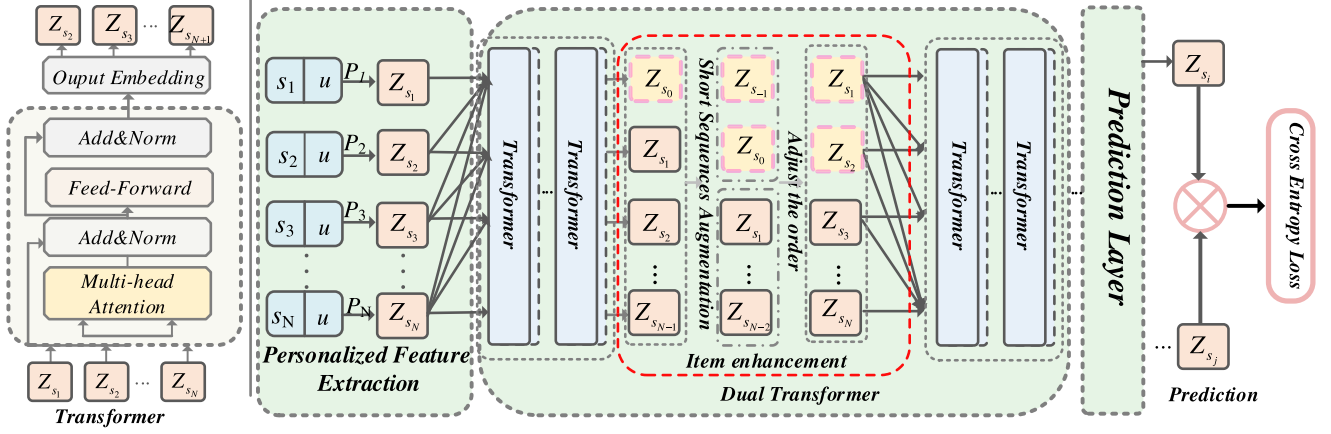


Fig. 2. An overview of the proposed Personalized Dual Transformer Network, consisting of three parts: (1) Personalized Feature Extraction is proposed to extract both user-specific features and historical behavior features. (2) Dual Transformer could retrieve a user's relevant historical behaviors and then mix them with the user's recent behaviors to form the final recommendation. (3) Prediction Layer could predict the next potentially exciting item.

given a user's action sequence S^u and a maximum sequence length N , the sequence is truncated by removing the earliest items if $|S^u| > N$. On the other hand, if $|S^u| < N$, the sequence is padded with zeros to achieve a fixed-length sequence $s = (s_1, s_2, \dots, s_N)$. The Transformer employs the scaled dot-product attention, as illustrated in Eq. (2).

$$A(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V, \quad (2)$$

where Q, K, V represent queries, keys and values respectively and $\sqrt{d}$ is the scaling factor used to prevent large values in the inner product. Furthermore, other components in Transformer will be introduced in the next section.

3.3. Personalized Feature Extraction Module

Advanced feature extraction from raw data is widely recognized as a crucial transformation in sequential recommendation applications [49–51]. To make our model personalized, we present two types of embeddings: item embeddings and user embeddings. We define an item embedding matrix $E^I \in R^{I \times d_i}$ and a user embedding matrix $E^U \in R^{U \times d_u}$, where d_i and d_u indicate the latent dimension of item and user respectively. We also define a learnable position embedding matrix $P \in R^{N \times d}$, where $d = d_u + d_i$. Its element P_v represents the position embedding for the v -th position within a sequence. Therefore, the input embedding E_s could be calculated as follows.

$$E_s = \begin{bmatrix} [E_{s_1}, E_u] + P_1, \dots, [E_{s_1}, E_u] + P_1, \dots, [E_{s_N}, E_u] + P_N \\ [Z_{s_1}, \dots, Z_{s_1}, \dots, Z_{s_N}] \end{bmatrix} \quad (3)$$

where $[E_{s_N}, E_u]$ represents concatenating item embedding $E_{s_N} \in R^{d_s}$ and user embedding $E_u \in R^{d_u}$ at N -th position in a sequence. N denotes the maximum length of the sequence. If the length of the user's historical behavior sequence longer than N , we take the most recent N items [9]. Through the Personalized Feature Extraction Module, we can capture the features of the user and the user's historical behaviors.

Since the Personalized Feature Extraction Module contains user embeddings and item embeddings, this will inevitably introduce a large number of parameters. To prevent overfitting, stochastic shared embedding regularization is employed to randomly replace an embedding with some predefined probability, which has the effect of regularizing the Personalized Feature Extraction Module. We use SSE probabilities p_u, p_i for input user/item embeddings. For the convenience of introduction, we take item embeddings as an example to explain. First, we generate a random number α , and then we do the following

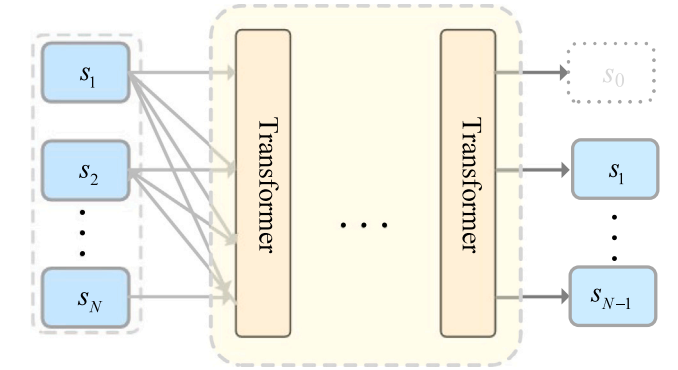


Fig. 3. The structure of Item enhancement module.

operations for any item embedding E_{s_i} of user u .

$$E_{s_i} = \begin{cases} E_{s_i}, & \alpha \leq p_i \\ \hat{E}_{s_i}, & \alpha > p_i \end{cases} \quad (4)$$

where $\hat{E}_{s_i}$ represents the random item embedding.

3.4. Dual Transformer

Given that Transformers can typically handle fixed-length sequences, excessive zero-padding may not be informative for predicting the next item, especially for users with short historical sequences [9,25,26]. Extracting their interests and providing accurate recommendations becomes challenging. Therefore, we propose a Dual Transformer to find related items to add the length of the user's historical sequences.

3.4.1. Item enhancement module

Through the Personalized Feature Extraction Module, we obtain personalized characteristics of the user and the characteristics of their interaction historical sequence. However, excessive zero-padding in the Personalized Feature Extraction Module may hinder the complete extraction of the user's interests. To tackle this problem, we design an Item enhancement module to retrieve items relevant to the user's historical behaviors. So the task in the Item enhancement module can be represented as $p(s_0|S^u)$. s_0 represents the previous item in a sequence. We mask the prior item on the left and force the Transformer to predict it. However, this raises two questions: 1. What is the required sequence length for item enhancement? 2. How many items need to be increased? Therefore, we define two parameters, L and H , where L represents the

threshold value for the sequence length, and H represents the number of enhanced items.

Similar to the Transformer architecture [9,21], the Item enhancement layer is constructed above the Personalized Feature Extraction Module by stacking multiple layers of self-attention to recursively produce pseudo-prior items. As illustrated in Fig. 3, the Item enhancement layer facilitates the retrieval of the prior item s_0 . Similarly, when the user's sequence length is shorter than L , we can obtain H items related to the historical sequence.

$$S^{u'} = [s_{-H+1}, s_{-H+2}, \dots, s_0]. \quad (5)$$

To preserve the sequence order, we place the related items $S^{u'}$ before the original sequence S^u to form a new sequence $[s_{-H+1}, s_{-H+2}, \dots, s_0, s_1, \dots, s_{N-H}]$. Through adjusting the order, we can get a new sequence $\hat{S}^u = [S^{u'}, S^u] = [s_1, s_2, \dots, s_N]$. Therefore, its corresponding embedding could be represented as follows.

$$\hat{E}_s = [\hat{Z}_{s_1}, \hat{Z}_{s_2}, \dots, \hat{Z}_{s_N}]. \quad (6)$$

As the transformer model requires a fixed length of input, we select only the most recent N items if the new sequence exceeds N in length, and zero-pad it if it is shorter than N .

3.4.2. Interest representation learning

Within this module, we leverage the historical sequence in conjunction with the enhanced items to extract the user's interests. To learn the user's interests, we use multiple self-attention blocks. Since the Transformer requires a fixed-length input, if the sequence (after item enhancement) exceeds the length of N , we consider only the last N items, a self-attention block encompasses two components: multi-head attention and feedforward network. The multi-head self-attention based on the Personalized Feature Extraction Module is denoted as follows.

$$Att(F^l) = [head_1, \dots, head_l, \dots, head_h]W^O, \quad (7)$$

$$head_i = A(F^l W_i^Q, F^l W_i^K, F^l W_i^V), \quad (8)$$

where W^O , W_i^Q , W_i^K and W_i^V are the corresponding learnable parameters and F^l is the input of the l -th layer. If $l = 0$, then $F^0 = \hat{E}_s$. The Attention function A is implemented by scaling the dot-product operation. While our enhancement layer is able to combine all items, user information and absolute locations, it achieves this through linear combinations. After each enhancement layer, we perform two linear transformations with a ReLU activation in the middle, as illustrated below.

$$F^l = [FFN(F_1^l)^T; \dots; FFN(F_i^l)^T; \dots; FFN(F_N^l)^T], \quad (9)$$

$$FFN(F_i^l) = (\text{ReLU}(F_i^l W_1 + b_1))W_2 + b_2, \quad (10)$$

where W_1 , W_2 , b_1 and b_2 are trainable parameters.

3.5. Prediction layer

Finally, the user's preference score for item i at the final self-attention layer can be computed as depicted in Eq. (11).

$$r_{it} = F_i^B N_i^t, \quad (11)$$

where F_i^B denotes the output hidden units associated with the Transformer at the last timestamp, B represents the number of self-attention blocks, and N_i^t denotes the embedding of the user and item. At time t , the prediction probability of the user for item i can be calculated as follows.

$$z_{it} = \sigma(r_{it}), \quad (12)$$

where σ represents the sigmoid function, r_{it} denotes the predicted score of item i by user at t .

Algorithm 1 The PDTN learning algorithm

Require:

Sequence: S^u ; User: u ; Epoch: e ; Batch-size: b ; Hyper-parameters: L , H , p_u , p_i .

Ensure:

Output: Network Parameters Θ .

- 1: Initialize the parameters of PDTN.
- 2: **for** $epoch = 1, 2, \dots, e$ **do**
- 3: **for** Assemble training samples: $\{u_i, S^u\}_{i=1}^b$ **do**
- 4: get E_s with Eq. (3);
- 5: use SSE probabilities p_u, p_i to randomly replace user embedding and item embedding with Eq. (4);
- 6: **if** the length of sequence shorter than L **then**
- 7: $p(S_0^u | S^u)$ to predict H potentially interesting items for users to get $S^{u'}$, by adjusting the order we get the corresponding embedding $\hat{E}_s$;
- 8: **end if**
- 9: Interest Representation Learning with Eq. (7), Eq. (8), Eq. (9);
- 10: Predict the next item with Eq. (11), Eq. (12);
- 11: Compute ℓ via Eq. (13).
- 12: $\Theta \leftarrow \text{Backward}(\ell)$.
- 13: **end for**
- 14: **end for**
- 15: **return** Network Parameters Θ .

We adopt binary cross entropy loss ℓ to measure the correctness of the next-item prediction, as described below.

$$\ell = \sum_{S^u} \sum_{i \in \{1, 2, \dots, n\}} [\log(z_{it}) + \log(1 - z_{jt})], \quad (13)$$

where z_{it} denotes the ground truth item and z_{jt} indicates the negative sampled item from items that user u never interacts with.

In our model, we utilize Stochastic Shared Embeddings, which involves randomly replacing an embedding with another embedding with a predefined probability during the SGD process. This technique serves as a regularization method for the Personalized Feature Extraction Module. Without SSE, existing well-known regularization techniques such as layer normalization, dropout, and weight decay are unable to effectively prevent severe overfitting when user embeddings are introduced. In the training process of the PDTN framework, we employ the Adam optimizer [55] with a learning rate of $lr = 0.001$. To enhance training efficiency, we employ mini-batch SGD. Algorithm 1 provides a concise overview of the complete training procedure.

4. Experiments

To evaluate the proposed Personalized Dual Transformer Network, we perform experiments on three standard benchmark datasets. Recent state-of-the-art methods on the sequential recommendation task are included for comparison. Furthermore, we design ablation studies to evaluate the impact of individual model components.

4.1. Experimental settings

Datasets. In this paper, we assess the effectiveness of our method using the Amazon review datasets [56] and the Yelp dataset. Specifically, we focus on three categories: Sports and Outdoors 5-core, Toys and Games 5-core, and Beauty 5-core, selected from the Amazon review datasets for our evaluation. Yelp dataset is a dataset for business recommendations. Due to its extensive size, we have specifically utilized transaction records from January 1st, 2019 onwards for our analysis. Detailed statistics about the datasets can be found in Table 2. In

addition, we divide the sequences into five groups according to the frequency of recent interactive items and show the aggregate of each group in Table 3, where we can see that most items are located in the lowest frequency group. For each user, we designate the last item in the behavior sequence as the test data. The items preceding the last item are allocated as the validation set, while the remaining items are utilized for training purposes.

Evaluation Metric. We utilize HR@K (HitRate), NDCG@K (Normalized Discounted Cumulative Gain), and MRR (Mean Reciprocal Rank) to evaluate the performance, where $K \in \{1, 5, 10\}$ [57]. HR@K denotes the proportion of accurately recommended items among the top K positions in the recommendation list and evaluates the accuracy of the recommendations by assessing how well the recommended items align with the user's interests. NDCG@K evaluates the significance of the recommended items by considering their ranking in the list. It assigns higher importance to items with greater relevance and assigns them to higher positions. MRR primarily evaluates the ability of the recommendations to capture the user's attention. It assesses the quality of the recommendation order by calculating the average reciprocal ranks of the recommended items in the list.

Baseline Methods. We compare the proposed PDTN with the following state-of-the-art sequential recommendation methods:

- **PopRec** recommends items to users based on their frequent appearance in the users' historical interaction data, assuming that the frequent appearance indicates user's preference for these items. The recommended items are then ranked by popularity before being presented to the user.
- **FM** [31] utilizes factorization models to capture and model the pairwise interactions between variables.
- **AutoInt** [58] uses a neural network architecture incorporating multi-head self-attention mechanism to capture the feature interactions.
- **GRU4Rec** [34] models the user's interests evolution process by utilizing a GRU (Gated Recurrent Unit) network to process the user's historical behavior sequence as input and then generates recommendations by predicting the next item that the user might be interested in.
- **Caser** [35] employs convolutional operations to encode the user's historical behavior sequence, integrates it with item embeddings, and subsequently computes similarity scores for each user-item pair to generate personalized recommendations.
- **SASRec** [9] utilizes a self-attention mechanism with multiple heads to encode the user's historical behavior sequence into a vector representation.
- **BERT4Rec** [27] takes the user's historical behavior sequences and item metadata as input, employs the pre-trained BERT model to learn contextualized embeddings for each position in the sequence, and generates next-item recommendations based on these embeddings.
- **S³-Rec** [59] leverages the inherent data correlation to acquire self-supervised signals, and employs pre-training methods to enhance data representation, thereby improving sequence recommendation.
- **DT4SR** [24] is a sequential recommendation method that utilizes distribution and uncertainty modeling to capture the uncertainty in the user's historical behavior sequences, aiming to achieve more accurate personalized recommendations.
- **SSE-PT** [25] proposes a personalized Transformer to improve recommended performance.
- **FMLP-Rec** [60] constructs the user's historical behavior sequence into a two-dimensional matrix input network, applies a series of vertical and horizontal convolutional filters on the matrix to capture semantic information at different levels, and uses a multi-head self-attention mechanism to enhance the ability of feature interactions.

Table 2

Statistics of datasets used in the paper.

Dataset	Users	Items	Interactions	Density
Sports	35,598	18,357	296,337	0.045%
Toys	19,412	11,924	167,597	0.072%
Beauty	22,363	12,101	198,502	0.073%
Yelp	30,431	20,033	316,354	0.05%

Table 3

Number of sequence items ending in different frequency groups.

Dataset	[= 3]	[3, 7]	[7, 20]	[≥ 20]
Sports	1146	16324	6739	1119
Toys	6579	8645	3380	808
Beauty	7162	10078	4104	1019

- **DualRec** [61] achieves the disentanglement of the past and the future and the mutual reinforcement of the past and the future through a new dual network.
- **SPMRec** [62] serves as a controllable substitute to address degeneration and strike a balance between recommendation diversity and performance.
- **BSARec** [63] utilizes the Fourier transform to incorporate an inductive bias by analyzing detailed sequential patterns and combining both low and high-frequency information to reduce over-smoothing.

Implementation Details. For AutoInt [58], GRU4Rec [34], Caser [35], BERT4Rec [27], FMLP-Rec [60], DualRec [61], SPMRec [62] and BSARec [63], we use the code provided by the author. For SASRec [9], S³-Rec [59], DT4SR [24] and SSE-PT [25], we implement them on the framework of TensorFlow 2.10.0. In the evaluation methodology, only S³-Rec considers item attribute information, while other methods do not utilize item attribute information. To ensure a fair comparison, we only retain the item prediction and segment prediction tasks from S³-Rec for model learning. For common hyperparameters in all models, we search the hidden dimension size d from {16, 32, 64, 128, 256}, the L_2 regularization from {0.0, 0.0001, 0.001, 0.01, 0.1}. For max input sequence length N , we search from {50, 100} for all SR methods. All other hyperparameters and initialization strategies either follow the recommendations of the method authors or are adjusted based on the validation set. We report the results of each baseline at its optimal hyperparameter settings.

For our proposed PDTN, we search the hidden dimension size d from {16, 32, 64, 128, 256}, we search the L_2 regularization from {0.0, 0.0001, 0.001, 0.01, 0.1}. We select the attention heads h from {1, 2, 4, 8}, we set the layer number $B = 2$, and a batch size b of 128. We set the maximum sequence length as 100, the threshold value of sequence length L from {6, 10, 14, 18} and the number of enhancing item H from {4, 8, 12, 16, 20}. All the models are trained on the NVIDIA RTX 3090 GPU.

4.2. Experimental results

To conduct a comprehensive evaluation and comparison, we initially calculate HR@1, HR@5, NDCG@5, HR@10, NDCG@10, and MRR for all baseline methods on three benchmark datasets to show the performance advantages of our proposed PDTN framework.

HR@1, HR@5, NDCG@5, HR@10, NDCG@10, and MRR scores for recommendation frameworks on four benchmark datasets (Sports and Outdoors 5-core dataset, Toys and Games 5-core dataset, Beauty 5-core dataset, and Yelp dataset) are listed in Tables 4–7, which indicates our proposed PDTN substantially outperforms other state-of-the-art in most cases. Based on the results, we can find that PopRec, FM, and AutoInt perform poorer than sequential recommendation models, which can explain the importance of mining the historical sequences of user behavior. When comparing sequential recommendation baseline models, we

Table 4

Performance comparison of different methods on the Sports dataset.

Methods	HR@1	HR@5	NDCG@5	HR@10	NDCG@10	MRR
PopRec	0.0763	0.2293	0.1538	0.3423	0.1902	0.1660
FM	0.0489	0.1603	0.1048	0.2491	0.1334	0.1202
AutoInt	0.0644	0.1982	0.1316	0.2967	0.1633	0.1435
GRU4Rec	0.1160	0.3055	0.2126	0.4299	0.2527	0.2197
Caser	0.1135	0.2866	0.2020	0.4014	0.2390	0.2100
SASRec	0.1474	0.3691	0.2616	0.4945	0.3021	0.2608
BERT4Rec	0.1255	0.3375	0.2341	0.4722	0.2775	0.2378
S ³ – Rec	0.1471	0.4346	0.2498	0.4644	0.2880	0.2529
SSE-PT	0.1502	0.3780	0.2676	0.5072	0.3092	0.2663
DT4SR	0.1541	0.3676	0.2644	0.4918	0.3044	0.2650
FMLP-Rec	0.1646	0.3803	0.2760	0.5059	0.3165	0.2763
DualRec	0.1947	0.4127	0.3080	0.5383	0.3485	0.3078
SPMRec	0.1820	0.3733	0.2824	0.4668	0.3126	0.2822
BSARec	0.2005	0.4108	0.3095	0.5273	0.3470	0.3089
PDTN	0.1891	0.4428	0.3201	0.5774	0.3636	0.3145

Table 5

Performance comparison of different methods on the Yelp dataset.

Methods	HR@1	HR@5	NDCG@5	HR@10	NDCG@10	MRR
PopRec	0.0801	0.2415	0.1622	0.3609	0.2007	0.1740
FM	0.0624	0.2036	0.1333	0.3153	0.1692	0.1470
AutoInt	0.0731	0.2249	0.1501	0.3367	0.1860	0.1616
GRU4Rec	0.2053	0.5437	0.3784	0.7625	0.4375	0.3630
Caser	0.2188	0.5111	0.3696	0.6661	0.4198	0.3595
SASRec	0.2375	0.5745	0.4113	0.7373	0.4642	0.3927
BERT4Rec	0.2405	0.5976	0.4252	0.7597	0.4778	0.4026
S ³ – Rec	0.2591	0.6085	0.4401	0.7725	0.4934	0.4190
SSE-PT	0.2380	0.5756	0.4129	0.7470	0.4655	0.4132
DT4SR	0.2390	0.5782	0.4135	0.7581	0.4915	0.4276
FMLP-Rec	0.2727	0.6191	0.4527	0.7720	0.5024	0.4299
DualRec	0.2893	0.6328	0.4681	0.7896	0.5190	0.4466
SPMRec	0.2483	0.6036	0.4325	0.7633	0.4844	0.4098
BSARec	0.2957	0.6335	0.4684	0.7816	0.5199	0.4465
PDTN	0.2896	0.6342	0.4693	0.7901	0.5211	0.4471

Table 6

Variant comparisons of our PDTN at HR@1, HR@5, NDCG@5, HR@10, NDCG@10, MRR on the Sports dataset.

Methods	HR@1	HR@5	NDCG@5	HR@10	NDCG@10	MRR
Transformer	0.1474	0.3691	0.2616	0.4945	0.3021	0.2608
PDTN-personalized module	0.1819	0.4420	0.3199	0.5770	0.3635	0.3122
PDTN-Dual Transformer	0.1502	0.3780	0.2676	0.5072	0.3092	0.2663
PDTN	0.1891	0.4428	0.3201	0.5774	0.3636	0.3145

Table 7

Variant comparisons of our PDTN at HR@1, HR@5, NDCG@5, HR@10, NDCG@10, MRR on the Toys dataset.

Methods	HR@1	HR@5	NDCG@5	HR@10	NDCG@10	MRR
Transformer	0.1474	0.3691	0.2616	0.4945	0.3021	0.2608
PDTN-personalized module	0.1986	0.4563	0.3346	0.5757	0.3734	0.3260
PDTN-Dual Transformer	0.1653	0.39177	0.2838	0.4941	0.3169	0.2787
PDTN	0.2035	0.4564	0.3352	0.5757	0.3738	0.3275

observe that SASRec and BERT4Rec outperform GRU4Rec and Caser. This is attributed to utilizing a Transformer-based encoder in SASRec and BERT4Rec. The superior performance of Transformer-based encoders suggests their greater suitability for modeling sequential data compared to RNN-based models (such as GRU4Rec) and CNN-based models (*i.e.*, Caser). In addition, S³ – Rec achieves better performance than SASRec and BERT4Rec, demonstrating that the self-supervised method effectively enhances the performance of sequential recommendation. Furthermore, we observe that SSE-PT, based on SASRec, outperforms SASRec. This suggests that Stochastic Shared Embedding effectively improves the performance of sequential recommendations. At the same time, DT4SR performs better than SASRec, which indicates modeling items and sequences as Elliptical Gaussian Distributions in the sequential recommendation is effective. However, compared with the most recent methods DualRec and BSARec, we found that our model is not as good as these two methods in terms of HR@1. This is because our model increases the length of the sequence by predicting some new

related items, which can simulate the user's purchase status to a certain extent. However, there will be some noise in this case, which makes it difficult for our model to reach a high level on HR@1. However, it is relatively good on fuzzy recommendation (HR@5, NDCG@5, HR@10, NDCG@10, and MRR). Therefore, our model is lower than DualRec and BSARec on evaluation HR@1 and higher than them on HR@5, NDCG@5, HR@10, NDCG@10, and MRR.

Finally, comparing with all the baselines, it is obvious that our approach PDTN performs consistently better than them by a wide margin on Sports, Toys, and Beauty datasets, and improves the performance of BSARec by on average 0.01% ~ 6.2% for different indicators. However, our method has not significantly improved on the Yelp dataset. This is because the statistical amount of data in the Yelp dataset is relatively large. Many users have amounts of similar items, making it difficult for the model to make choices. The user features learned by the model are not so distinctive, making it difficult to make correct choices in some judgments. Therefore, the model performance is rarely improved. The

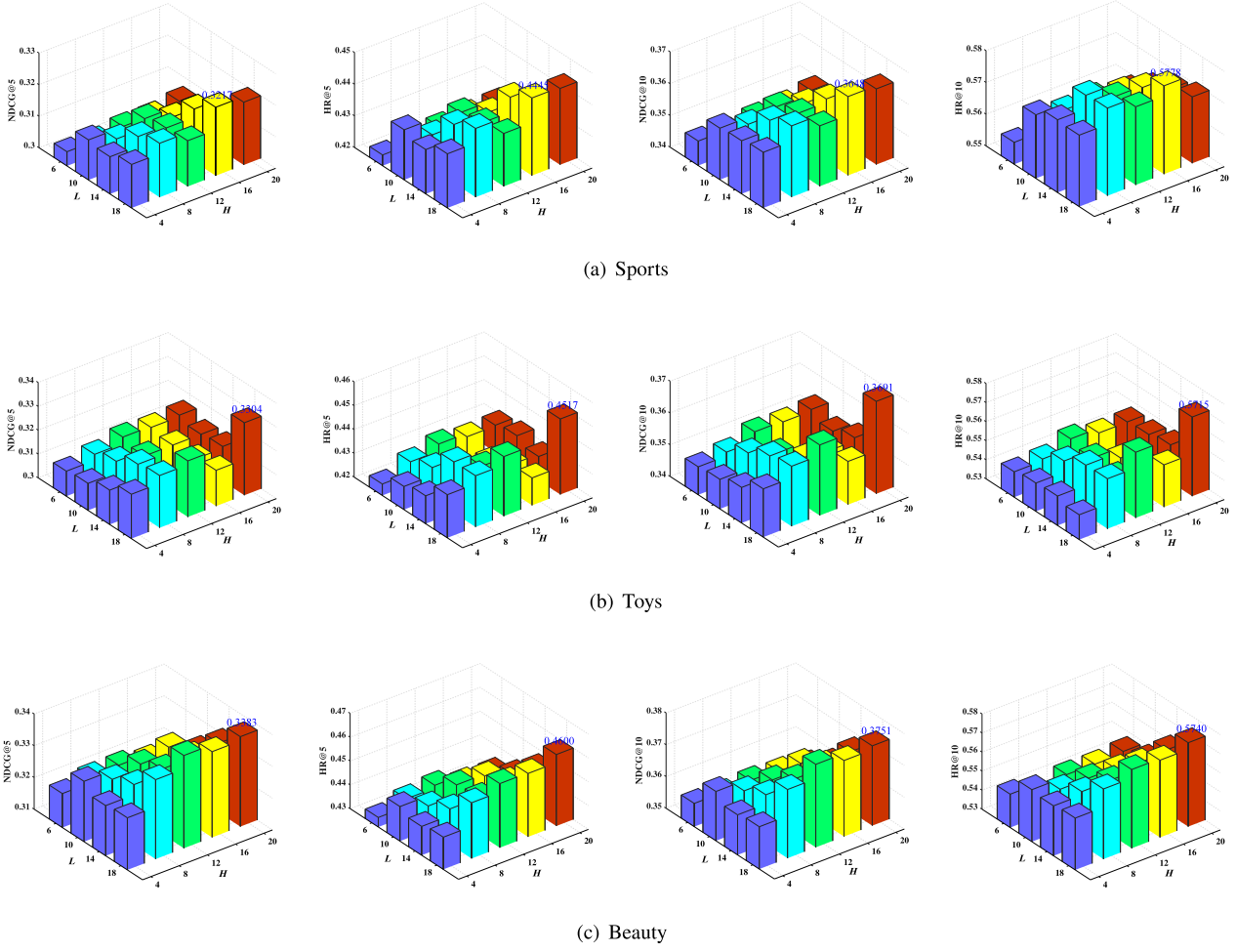


Fig. 4. Performance of different threshold for item enhancement L and the number of item enhancement H on Sports, Toys and Beauty dataset w.r.t. NDCG@5, HR@5, NDCG@10 and HR@10.

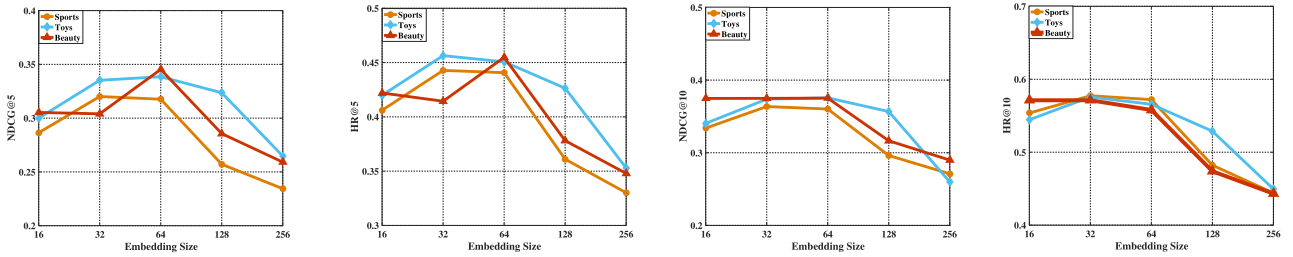


Fig. 5. Effect of the hidden dimensionality d on HR@5, NDCG@5, HR@10 and NDCG@10 for PDTN.

latest model BSARec also shows little improvement on this dataset. The difference with all these baselines is that our approach PDTN incorporates a personalized feature extraction module extracting advanced features, incorporating personalization, and utilizing Dual Transformer to address the issue of short sequences with zero-padding. As a result, the effectiveness of our PDTN approach is demonstrated.

4.3. Parameter analysis

To explore the influence of different parameters on the experimental results, we carry out a set of experiments using three benchmark datasets, the Sports dataset, the Toys dataset, and the Beauty dataset. The proposed model PDTN has two parameters, L and H , where L

denotes the threshold for item enhancement and H denotes the number of item enhancement, respectively. We use grid search to determine the appropriate parameter selection, and use a three-dimensional bar graph to show the experimental results of the combination of parameters L and H . In addition, to better show the best experimental results in each figure, we mark the corresponding results on the corresponding bars. With fixed other hyperparameters and Settings, we select the L from $\{6, 10, 14, 18\}$ and H from $\{4, 8, 12, 16, 20\}$. Fig. 4 shows the performance w.r.t. NDCG@5, HR@5, NDCG@10, and HR@10 of the default PDTN on three datasets. As observed across all datasets, performance consistently improves with an increase in the value of L , thereby confirming the significance of item enhancements. Furthermore, performance demonstrates further improvement when both L and H are increased simultaneously. As we can see, for the dataset of Sports, when

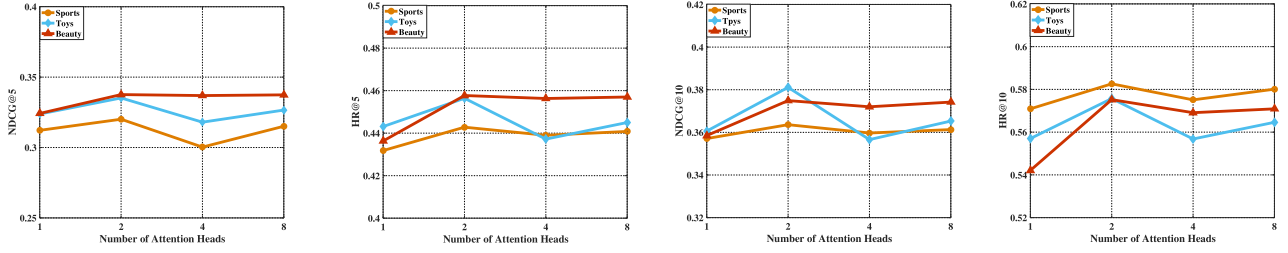


Fig. 6. Effect of Attention Heads on HR@5, NDCG@5, HR@10 and NDCG@10 for PD TN.

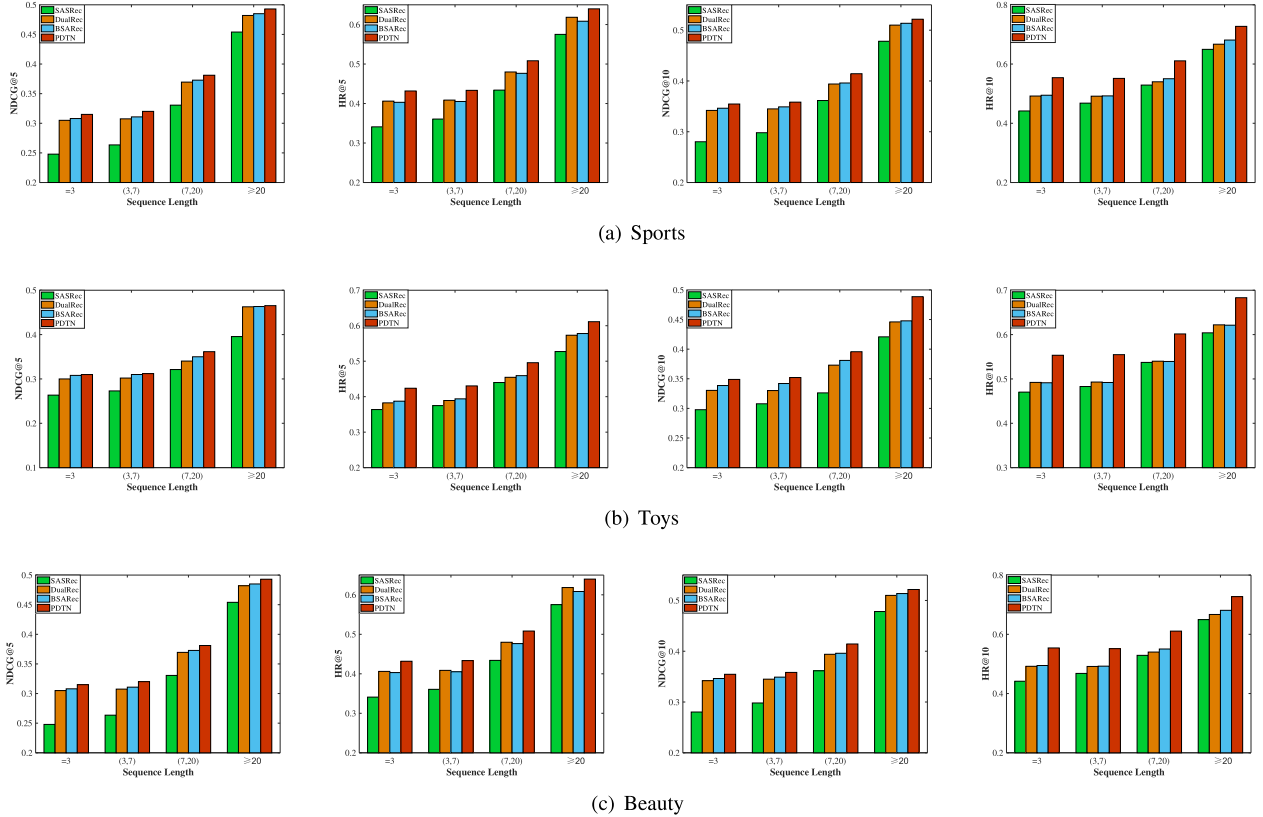


Fig. 7. Performance of different sequence lengths on Sports, Toys and Beauty dataset w.r.t. NDCG@5, HR@5, NDCG@10 and HR@10.

H equals 16 and L equals 18, the best results can be obtained. For the dataset of Toys, when H equals 20 and L equals 18, the best results can be obtained. Likewise, for the dataset of Beauty, when H equals 20 and L equals 18, the best results can be obtained.

4.4. Improvements analysis

Impact of Dimension Size d . To investigate the impact of Dimension Size d on the experimental results, we conduct a series of experiments using three benchmark datasets. Fig. 5 shows the effect of attribute embedding size d varying from 16 to 256 on the performance of PD TN. We can observe that by holding the other optimal hyperparameters constant, with the increase of embedding size, the performance of the model improves steadily. However, when the embedding size reaches a certain level, a higher embedding size may not make performance better. Considering that our model can achieve better performance when $d \geq 32$, we only report the results when $d = 32$ in the following analysis.

Impact of Attention Heads. To investigate the impact of Attention Heads on the experimental results, we conduct a series of experiments using three benchmark datasets. Fig. 6 shows the effect of attribute Attention Heads. Attention Heads varying from 1 to 8 on the performance of PD TN. We can observe that by holding the other optimal hyperparameters constant, with the increase of Attention Heads, the performance of the model improves steadily. When the value of Attention Heads equals 2, PD TN achieves the best performance on all datasets. We only report the results when the value of Attention Heads equals 2 in the following analysis.

4.5. Performances w.r.t sequence lengths

We divide the user sequences into 5 groups based on their length as shown in Table 3 and illustrate the performance of PD TN and other models on the Sports, Toys, and Beauty datasets in Fig. 7. We choose several models that are most similar to ours for comparison including SASRec, DualRec, and BSARec. The results show that PD TN consistently exhibits good performance w.r.t. NDCG@5, HR@5, NDCG@10, and

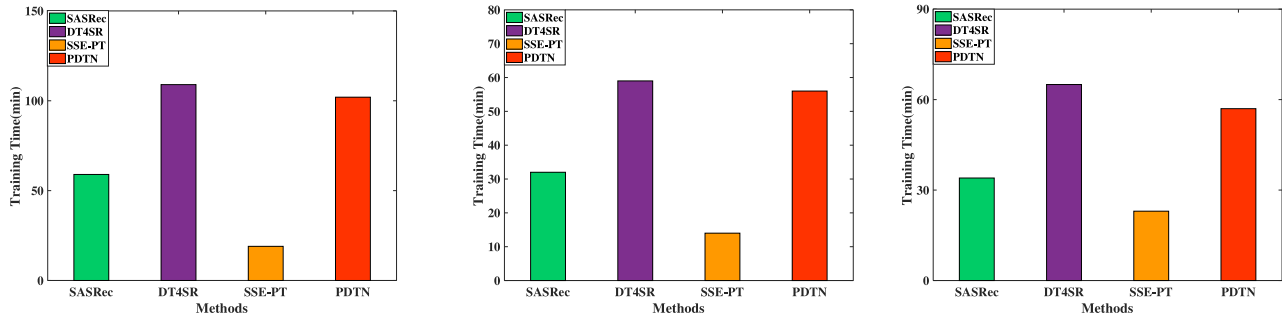


Fig. 8. Comparison of training time of different methods on Sports, Toys and Beauty datasets.

Table 8

Variant comparisons of our PDTN at HR@1, HR@5, NDCG@5, HR@10, NDCG@10, MRR on the Beauty dataset.

Methods	HR@1	HR@5	NDCG@5	HR@10	NDCG@10	MRR
Transformer	0.1643	0.3849	0.2792	0.4928	0.3141	0.2759
PDTN-personalized module	0.2069	0.4422	0.3319	0.5466	0.3657	0.3248
PDTN-Dual Transformer	0.1753	0.3685	0.2852	0.4877	0.3184	0.2830
PDTN	0.2071	0.4561	0.3376	0.5826	0.3749	0.3299

HR@10 across sequences of varying lengths, highlighting the effectiveness of our model. Secondly, PDTN greatly improves the performance of short sequences (that is, user sequences with fewer than 20 interactions). Additionally, this demonstrates the effectiveness of dual Transformer and personalized feature extraction modules.

4.6. Training time analysis

We evaluate the training time cost of several Transformer frameworks, including SASRec, DT4SR, and SSE-PT. Fig. 8 shows that the training time of our method is longer than SASRec and SSE-PT, its complexity does not affect the recommendation performance. At the same time, the training time of PDTN is shorter than DT4SR. DT4SR models the sequences as distributions and employs the Gaussian distance to measure the distance between distributions. The effectiveness of our framework in generating relevant historical behaviors in turn improves the effectiveness of recommendations.

4.7. Ablation experiment

In this section, we conduct ablation experiments on key components of PDTN to understand their effectiveness, including a personalized feature extraction module and a Dual Transformer. Table 6, Table 7, and Table 8 show the performance w.r.t. HR@1, HR@5, NDCG@5, HR@10, NDCG@10 and MRR of the default PDTN and other variants. The PDTN with all components. We develop the variants by substituting some components, which are Transformer, PDTN-personalized feature extraction module, and PDTN-dual transformer.

Effectiveness of the personalized feature extraction module In our Transformer-based sequential approach, the personalized feature extraction module could extract both user and historical sequence features. To justify the effectiveness of the personalized feature extraction module, we compare the results of HR@1, HR@5, NDCG@5, HR@10, NDCG@10, and MRR between PDTN and traditional Transformer. The results of the three benchmarks are shown in Table 6, Table 7, and Table 8, it is obvious that compared with PDTN-Dual Transformer, our PDTN has improved by at least 2.8% on all evaluation metrics, especially on HR@5, with a notable improvement of 9.49%. This is because our personalized feature extraction Module in PDTN extracts the user's personalized information, thereby enhancing the recommendation performance.

Effectiveness of the Dual Transformer We have demonstrated the effectiveness of the personalized feature extraction module. The

Dual Transformer could retrieve a user's relevant historical behaviors to increase the length of the short sequence. We compare the results of HR@1, HR@5, NDCG@5, HR@10, NDCG@10 and MRR between PDTN and traditional Transformer. The results on the three benchmarks are shown in Table 6, Table 7, and Table 8, it is obvious that compared with PDTN-personalized module, our PDTN absolute advantages in all evaluation metrics, especially on HR@5, with a notable improvement of 1.39%. This is because the Dual Transformer can improve performance, which verifies that strengthening the item sequence to comprehensively extract the user's interests is helpful.

5. Conclusion and future work

This paper proposes a Dual Transform sequence recommendation method called Personalized Dual Transformer Network for Sequential Recommendation. PDTN contains three main modules: Personalized Feature Extraction Module, Dual Transformer, and Prediction layer. The personalized feature extraction module can extract the user's features and the user's historical interaction to reflect the user's interests. At the same time, we introduce SSE regularization to prevent overfitting. For the Dual Transformer module, a Transformer is applied to find related items of user history behavior for projecting enhancement to add relevance, and then mix it with the user's recent history into a Transformer to predict the next item. For the prediction layer, an MF layer is used to predict the relevant items. Extensive experiments are conducted on four datasets, the results demonstrate the effectiveness of PDTN and also demonstrate its superiority in mitigating the cold-start project recommendation problem.

Although our proposed Transformer-based recommendation model achieves good performance, users' real-world sequential behavior exhibits uncertainty rather than determinism, which presents a significant challenge to the current dot product-based self-attention mechanism. In the future, we plan to enhance the transformer architecture by incorporating uncertainty into model training.

CRedit authorship contribution statement

Meiling Ge: Writing – review & editing, Writing – original draft, Methodology. **Chengduan Wang:** Supervision. **Xueyang Qin:** Formal analysis. **Jiangyan Dai:** Formal analysis. **Lei Huang:** Software. **Qibing Qin:** Writing – review & editing. **Wenfeng Zhang:** Software.

Declaration of competing interest

The manuscript has not been published before and is not being considered for publication elsewhere. All authors have contributed to the creation of this manuscript for important intellectual content and read and approved the final manuscript. We declare that we have no conflicts of interest to this work.

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Data availability

Data will be made available on request.

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- Meiling Ge** received the master's degree with the College of Computer Science, Shaanxi Normal University. She is currently a teaching assistant with the School of Computer Engineering, Weifang University, Weifang, China. Her current research interests include recommendation, multimedia content analysis and retrieval, machine learning, and data mining.
- Chengduan Wang** is currently a Professor with the School of Computer Engineering, Weifang University, Weifang, China. His current research interests include information processing, machine learning, and data mining.
- XueYang Qin** received his Ph.D. degree in computer software and theory from Dalian University of Technology, Dalian, in 2024, and received his M.S. degree in computer systems organization from Shaanxi Normal University, Xian, in 2019. His main research interests include cross-modal retrieval, information extraction, and multimodal data processing.
- Jiangyan Dai** is currently an associate professor with the School of Computer Engineering, Weifang University, China. She received the Ph.D. degree from the School of Mathematics and Statistics, Northeast Normal University, in 2014. Her research interests are image processing, computer vision, and machine learning.
- Lei Huang** is currently a professor with the Ocean University of China, Qingdao, China. He received the Ph.D. degree in computer science from the Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China, in 2013. His current research interests are in the fields of multimedia content analysis and retrieval, computer vision, pattern recognition and machine learning.
- Qibing Qin** is currently an associate professor with the School of Computer Engineering, Weifang University, Weifang, China, and is an Academic Visitor of the Faculty of Information Science and Engineering, Ocean University of China, Qingdao. He received the Ph.D. degree in computer science from Ocean University of China, Qingdao, China, in 2021. His current research interests are in the fields of multimedia content analysis and retrieval, image processing, machine learning, and data mining.
- Wenfeng Zhang** is currently a lecturer with the College of Computer and Information Science, Chongqing Normal University, Chongqing, China, and is with the Lishui Institute of Hangzhou Dianzi University, Lishui, China. He received the Ph.D. degree in computer science from Ocean University of China, Qingdao, China, in 2021. His research interests include computer vision and machine learning.